



Probability Distributions - Solutions

Exercise 3.1. (a) Let X be the denomination of coins put into a car parking meter. It is known that X follows the following probability distribution:

x	1p	2p	5p	10p	20p	50p	£1	£2
$P(X = x)$	0.02	0.02	0.05	0.07	0.08	0.16	0.4	0.2

- (i) What is $P(X \geq 10p)$?
 (ii) What is $P(5p < X \leq 50p)$?
 (iii) What is $P(X \leq 25p)$?

(i) For $P(X \geq 10p)$, we want the probability that X takes a value of at least 10p. That includes 10p, 20p, 50p, £1 and £2. So the probability is $0.07 + 0.08 + 0.16 + 0.4 + 0.2 = 0.91$.

(ii) For $P(5p < X \leq 50p)$, we want the probability of X being 10p, 20p or 50p. So the probability is $0.07 + 0.08 + 0.16 = 0.31$.

(iii) For $P(X \leq 25p)$, we want X to be 1p, 2p, 5p, 10p or 20p. So the probability is $0.02 + 0.02 + 0.05 + 0.07 + 0.08 = 0.24$.

(b) A random variable has the following distribution.

x	1	2	3	4
$P(X = x)$	42%	18%	14%	?

What is $P(X = 4)$? Since the total of the probabilities must be 1, we have that $P(X = 4) = 100\% - (42\% + 18\% + 14\%) = 26\%$.

Exercise 3.2. (a) The discrete random variable X has the following probability distribution:

x	-100	-50	-20	-10	0	100	200
$P(X = x)$	0.19	0.05	0.24	0.01	0.1	0.001	0.409

- (i) Find the probability that the random variable takes a negative value. **0.49**
 (ii) Find the expectation of X . **55.5**
 (b) For the two probability distributions given in Exercise 3.1, find the mean of each.

$$(a) \sum xP(X = x) = (1 \times 0.02) + (2 \times 0.02) + (5 \times 0.05) + (10 \times 0.07) + (20 \times 0.08) + (50 \times 0.16) + (100 \times 0.4) + (200 \times 0.2) = 90.61$$

$$(b) \sum xP(X = x) = (1 \times 0.42) + (2 \times 0.18) + (3 \times 0.14) + (4 \times 0.26) = 2.24$$

Exercise 3.3. For the probability distributions given in Exercise 3.1 and 3.2, find the variance.

For 3.1:

$$(a) \sum (x - \mu)^2 P(X = x) = ((1 - 90.61)^2 \times 0.02) + ((2 - 90.61)^2 \times 0.02) + ((5 - 90.61)^2 \times 0.05) + ((10 - 90.61)^2 \times 0.07) + ((20 - 90.61)^2 \times 0.08) + ((50 - 90.61)^2 \times 0.16) + ((100 - 90.61)^2 \times 0.4) + ((200 - 90.61)^2 \times 0.2) = 4230.18$$

$$(b) \sum (x - \mu)^2 P(X = x) = ((1 - 2.24)^2 \times 0.42) + ((2 - 2.24)^2 \times 0.18) + ((3 - 2.24)^2 \times 0.14) + ((4 - 2.24)^2 \times 0.26) = 1.54$$

For 3.2:

$$\sum (x - \mu)^2 P(X = x) = ((-100 - 55.5)^2 \times 0.19) + ((-50 - 55.5)^2 \times 0.05) + ((-20 - 55.5)^2 \times 0.24) + ((-10 - 55.5)^2 \times 0.01) + ((0 - 55.5)^2 \times 0.1) + ((100 - 55.5)^2 \times 0.001) + ((200 - 55.5)^2 \times 0.409) = 15411.75$$

Exercise 3.4.

- (a) For each of the situations below, state which of the conditions of being modelled using a binomial distribution it fails to satisfy.
- (i) A coin is flipped until it lands "tails". This is not binomial - there isn't a fixed number of trials, so the third condition is violated.
 - (ii) An archery student fires 13 arrows, and the number of times they hit the target is recorded. This might not be binomial. It is reasonable to think that an archery student might get better as they shoot arrows. This means the probability of hitting the target may increase with each shot, so the probability is not the same each time. This violates the fourth condition.
 - (iii) A group of 15 friends arrive at the college, and the number of students who get to class late is recorded. This might not be binomial - it is likely that friends will stick together and if one person arrives late, it may influence the other students to wait for them. This means the students are not independent of each other, so the second condition is violated.
 - (iv) A die is rolled and the score on the die is recorded. This is not binomial - each roll of the die can give the score 1,2,3,4,5 or 6. This means there are more than 2 outcomes, so the first condition is violated.
- (b) Write down the expression for the binomial distribution for each of the examples below, by determining the values of n and p in each case.
- (i) In a class of 30 students, it is expected that 22% of them will have brown eyes. $B(30, 0.22)$
 - (ii) In one week, each day from Monday to Friday, Noel catches a train from Piccadilly station. He claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day. $B(5, 0.4)$

Exercise 3.5.

1. Let X be the distribution of the heights of men from Albania. We assume that X is a normal distribution, with a mean of 174cm and standard deviation of 11 cm. What is the probability that if we spoke to a random Albanian man, he will be taller than 160cm?

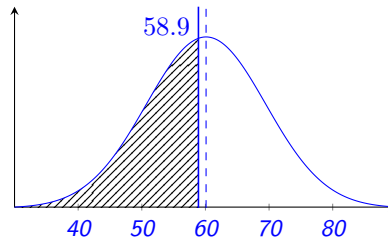
We can write $X \sim N(174, 11^2)$. We are looking for $P(X > 160)$. Find the z -score:

$$z = \frac{160 - 174}{11} = \frac{-14}{11} = -1.27$$

Since this is negative, we need to find the probability for $z = 1.27$ (the area to the left of the equivalent value on the opposite side). The table gives this as 0.8980, so the probability is 89.8%, meaning 89.8% of men in Albania are taller than 160cm.

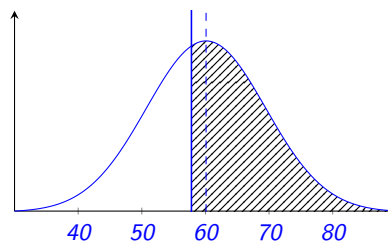
2. Let $X \sim N(60.1, 9.39^2)$.

(i) Find $P(X < 58.9)$.



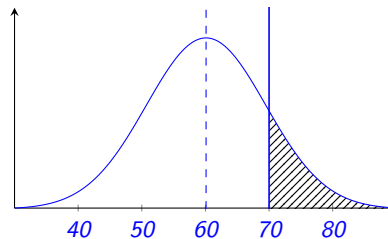
$z = \frac{58.9 - 60.1}{9.39} = \frac{-1.2}{9.39} = -0.13$ Looking up positive 0.13 in the table gives 0.5517, which is equal to the area to the right of 58.9. We want the other side, so $1 - 0.5517 = 0.4483$.

(ii) Confirm that $P(X > 57.8)$ is 0.5948.



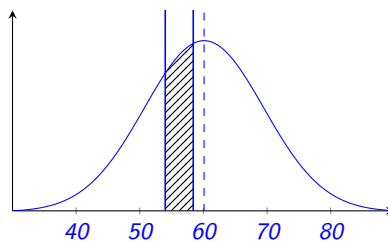
$z = \frac{57.8 - 60.1}{9.39} = \frac{-2.3}{9.39} = -0.24$ Looking up positive 0.24 in the table gives 0.5948, which is the area we need, so $P(X > 57.8) = 0.5948$ as required.

(iii) Find $P(X \geq 70)$.



$z = \frac{70 - 60.1}{9.39} = \frac{9.79}{9.39} = 1.04$ Looking up 1.04 in the table gives 0.8508, which is the area to the left of 70; we need the area to the right, so $1 - 0.8508 = 0.1492$.

(iv) Confirm that $P(54 \leq X < 58.4)$ is 0.1708.

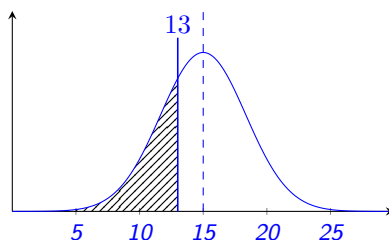


For 54, $z = \frac{54 - 60.1}{9.39} = -0.65$. For 58.4, $z = \frac{58.4 - 60.1}{9.39} = -0.18$.

Looking up the positive equivalents of these values in the table gives us that the area to the left of 54 is 0.7422, and the area to the left of 58.4 is 0.5714; we need the difference between these, so $0.7422 - 0.5714 = 0.1708$, as required.

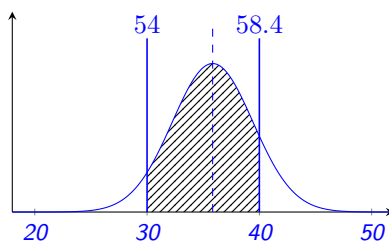
Exercise 3.6.

(a) Let $X \sim B(60, 0.25)$. Using the normal approximation to the binomial, find $P(X \leq 13)$.



Since $n > 30$, we can approximate $B(60, 0.25)$ by $N(15, 11.25)$. Since $np = 15 > 5$, we need to correct for continuity, so we find $P(X < 13.5)$. We need to calculate $z = \frac{13.5 - 15}{\sqrt{11.25}} = -0.45$. Looking up the positive equivalent of this values in the table gives us that the area to the right of 13 is 0.6736, so we need $1 - 0.6736 = 0.3264$.

(b) Let $X \sim B(56, 0.64)$. Using the normal approximation to the binomial, find $P(30 < X < 40)$.



Since $n > 30$, we can approximate $B(56, 0.64)$ by $N(35.84, 12.9)$. Since $np = 35.84 > 5$, we need to

correct for continuity, and find $P(30.5 < X < 39.5)$. For 30.5, $z = \frac{30.5 - 35.84}{\sqrt{12.9}} = -1.49$ and looking up the positive equivalent of this values in the table gives us that the area to the right of 29.5 is 0.9319, so we need $1 - 0.9319 = 0.0681$. For 39.5, we have $z = \frac{39.5 - 35.84}{\sqrt{12.9}} = 1.02$, which means the area to the left of 39.5 is 0.8461. We need the difference between the two areas, so $0.8461 - 0.0681 = 0.778$.